Lavkush Kumar

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PROFESSIONAL SUMMARY

Platform Engineer specializing in ML Platform and AI Infrastructure with **2.5+ years** of experience operating high-availability Kubernetes platforms, scalable data streaming, and CI/CD pipelines. Proficient in **Kubernetes (kubeadm, EKS), Terraform, Ansible, Jenkins, Prometheus, and Apache Kafka**. Track record of automating bare-metal clusters (**83% faster bootstrap**), stateful data backbones (**93% RTO reduction**), and open-source AI agent security control planes in **Rust** and **Python**.

TECHNICAL SKILLS

Platform & Orchestration: Kubernetes (kubeadm, AWS EKS), Docker, containerd, Helm, ArgoCD (GitOps), Harbor Registry
Infrastructure as Code: Terraform (modular architecture, remote state), Ansible (40+ roles, Jinja2), Bash, YAML
CI/CD & Security: Jenkins (declarative), GitLab CI, GitHub Actions, Trivy container scanning, SonarQube, RBAC
Observability & SRE: Prometheus, Grafana, Alertmanager, ELK Stack (Elasticsearch, Logstash Grok, Kibana), Velero DR
Data & Distributed Systems: Apache Kafka, Redis, PostgreSQL, TimescaleDB, MongoDB, pgvector (HNSW), MinIO S3
Languages & Frameworks: Python (FastAPI, AsyncIO, PyTest), Rust, SQL, Jinja2, RESTful APIs, WebSockets
ML & AI Infrastructure: MLflow (Model Registry, Lineage), Model Serving APIs, Vector Indexing, Cedar Policy Engine

PROFESSIONAL EXPERIENCE

CGI India (Client: PNC Bank GCC) — Jun 2026 – Present
DevOps Engineer (Contractor — Cloud Infra & Kubernetes) — *Hyderabad, India*

- **Provisioned** multi-tenant AWS EKS cluster infrastructure using Terraform modules and Ansible, establishing reproducible VPC peering, IAM roles, and auto-scaling node groups across environments.
- **Engineered** declarative Jenkins and GitLab CI pipelines with Trivy container scanning and automated quality gates, cutting deployment lead time by **40%**.
- **Hardened** ingress routing using Nginx Ingress and Kong Gateway with rate-limiting and mTLS, slashing gateway latency overhead by **15%** across **8** production microservices.

HFCL Limited — Aug 2025 – Jun 2026
Engineer 1 — DevOps / Platform Engineering (Platform Automation & Reliability) — *Bangalore, India*

- **Automated** bare-metal Kubernetes (kubeadm) cluster lifecycle via 40+ Ansible roles and Jinja2 templates (OS hardening, CNI, Harbor, Helm), cutting bootstrap time **from 2 days to 4 hours (83% reduction)**.
- **Operated** a 50-node on-premise Kubernetes platform for 15+ mission-critical microservices with namespace isolation, RBAC, and HPA, sustaining **99.95% platform availability**.
- **Orchestrated** stateful cluster workloads (Kafka, Redis, PostgreSQL, TimescaleDB, MongoDB) on Kubernetes via StatefulSets; configured Velero DR and Prometheus/Grafana alerts, slashing **RTO from 8h to 30min (93%)** and **MTTR by 45%**.
- **Standardized** declarative Jenkins pipelines with Docker, Helm, and Trivy security gates, enabling zero-downtime rolling updates and doubling release frequency (**2x**).

CGI India — Sep 2024 – Aug 2025
Security Analyst Apprentice (Log Analytics & Observability) — *Hyderabad, India*

- **Deployed** centralized log pipelines via the ELK Stack (Elasticsearch, Logstash, Kibana), building operational dashboards and query filters that reduced root-cause investigation time by **35%**.
- **Authored** custom Logstash Grok patterns to parse unstructured syslog and network telemetry into structured JSON schemas, accelerating incident triage by **50%**.

Stipe Solutions — Mar 2024 – Sep 2024
Full Stack Developer Intern (Mobile & Web Engineering) — *Bangalore, India*

- **Developed** 30+ production mobile screens and UI components using React Native and Flutter, integrating RESTful APIs and state management for **1,000+ daily active users**.

KEY PROJECTS

AegisAgent — Zero-Trust AI Agent Security Gateway | *Rust, Cedar Policy Engine, Kubernetes, Helm, Docker* **2026**

- **Architected** an open-source zero-trust security gateway for AI agents in Rust, integrating the Cedar Policy Engine to enforce ABAC over MCP tool invocations with **sub-75ms policy evaluation**.
- **Engineered** production Kubernetes Helm charts (aegis-gateway), ServiceMonitors, and NetworkPolicies, containerizing agent cages with reproducible GitHub Actions CI/SAST pipelines ensuring **100%** scan coverage.

Project Vulcan — Enterprise Automation Control Plane | *Python, FastAPI, pgvector, Redis Redlock, MinIO S3, WebSockets* **2025**

- **Architected** an infrastructure control plane in Python (FastAPI), implementing distributed locking via Redis Redlock and decoupled MinIO S3 storage to eliminate runbook race conditions.
- **Integrated** semantic incident retrieval using PostgreSQL with pgvector (HNSW indexing) and WebSocket live log streaming, cutting SRE incident diagnostics down to **under 2 minutes** per task.

EDUCATION

World College of Technology and Management — **Gurgaon, India**
B.Tech in Computer Science & Engineering — *Jun 2020 – May 2023*

Government Polytechnic, Patna-7 — **Patna, India**
Diploma in Computer Science Engineering — *Jul 2016 – Jun 2019*

CERTIFICATIONS

- **Microsoft Azure Fundamentals (AZ-900)** (2025) | **Microsoft Security Operations Analyst (SC-200)** (2025) | **Certified Kubernetes Administrator (CKA)** — *CNCF (In Progress, Expected 2026)*